



**AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification
Capstone Projects**

**Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator**

**Configure VPC using Public & private
Subnet in AWS**



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Objective

A **VPC (Virtual Private Cloud)** in Amazon Web Services (AWS) is a **logically isolated virtual network** inside AWS where you can launch and manage your cloud resources securely.

- Amazon Virtual Private Cloud (**VPC**) is a service that lets you launch **AWS** resources in a logically isolated virtual network that you define.
- It is logically isolated from other virtual networks in the AWS Cloud.
- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud.
- Amazon Virtual Private Cloud (Amazon VPC) hides the complexity, and simplifies the deployment of secure private networks.

Subnets:

A subnet is a sub range of IP addresses in the VPC. AWS resources can be launched into a specified subnet.

- After creating a VPC, you can add one or more subnets in each Availability Zone.



- ✓ *The public subnet* resources that must be connected to the internet.
- ✓ *The private subnet* resources that must remain isolated from the internet.

Internet Gateway:

An *internet gateway* is a VPC component that allows communication between resources in your VPC and the internet. It is horizontally scaled, redundant, and highly available. An internet gateway supports IPv4 and IPv6 traffic. An internet gateway serves two purposes. First, it provides a target in your VPC route tables for internet-routable traffic. Second, the internet gateway performs network address translation (NAT) for instances that were assigned public IPv4 addresses.

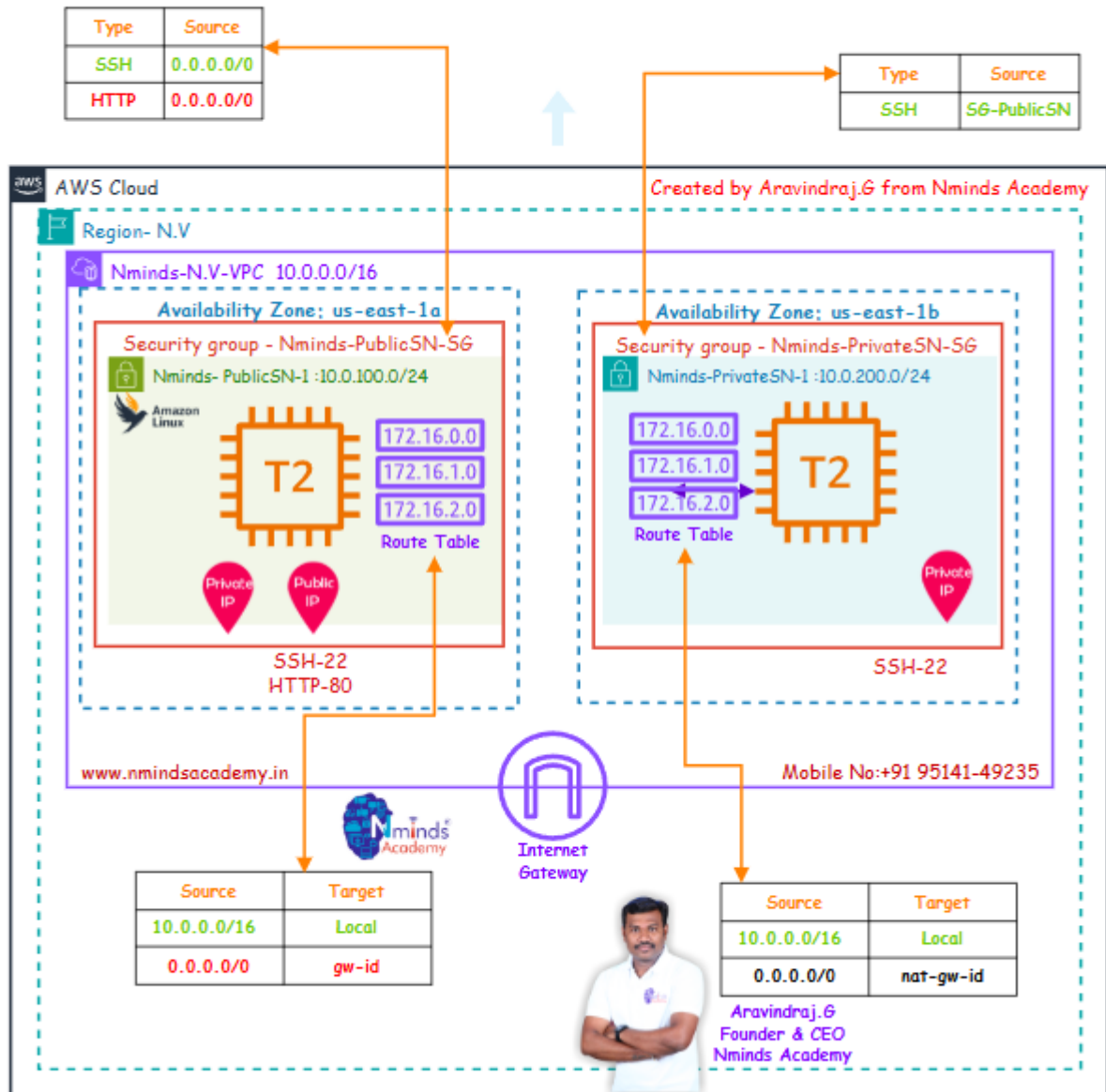


Internet gateway



Topology

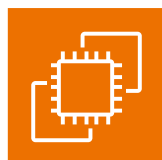
How to create custom VPC with Public Subnet and Private Subnet in AWS



Used Services in AWS



Amazon VPC



Amazon Elastic Compute Cloud
(Amazon EC2)



Apache Webserver



Internet gateway



Route table



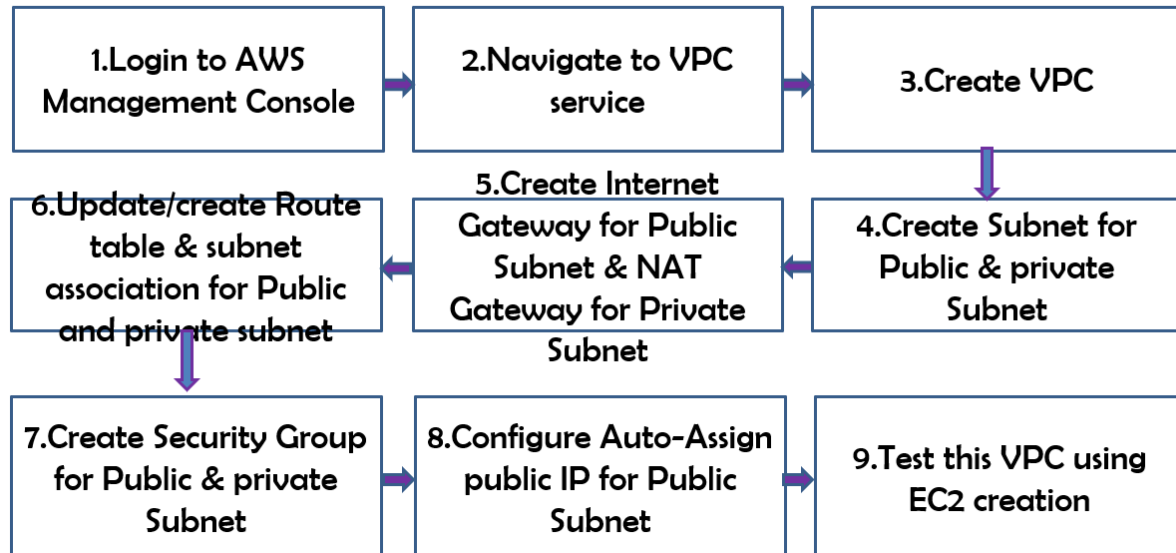
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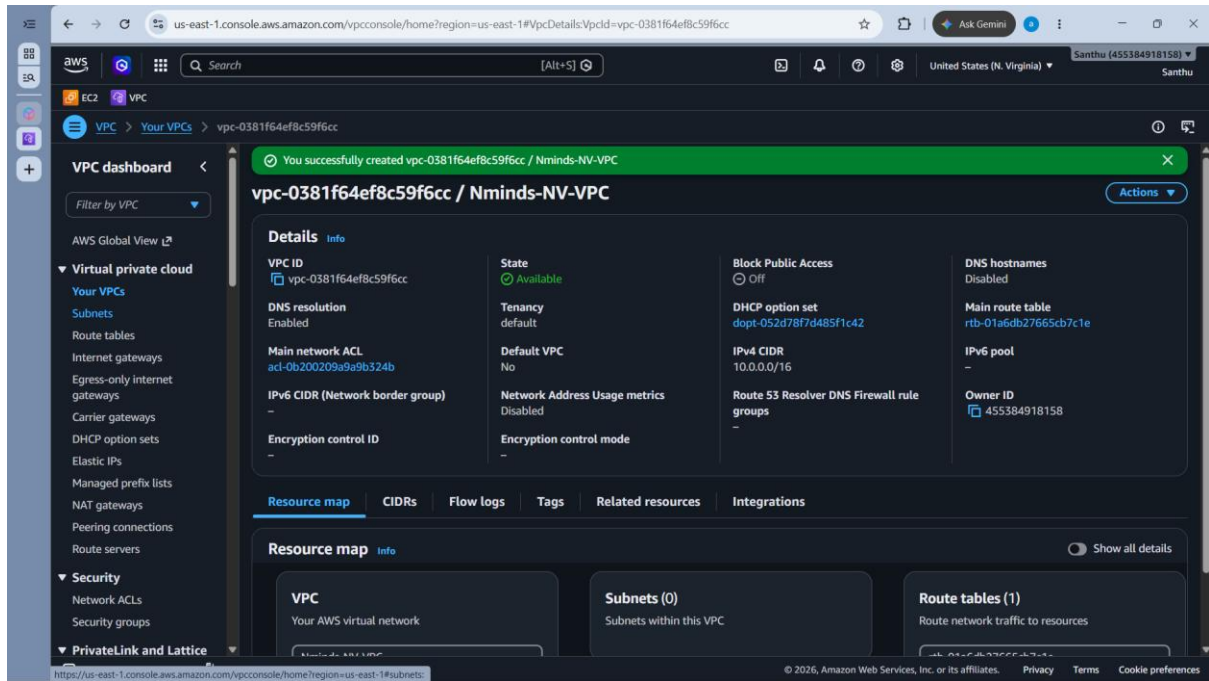
Execution Steps:

Steps to Configure Custom VPC in AWS



Execution Tasks:

Task1: Create the custom VPC with Public Subnet & private subnet based on the above topology:



us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSubnet:VpcId=vpc-0381f64ef8c59f6cc

Search [Alt+S]

United States (N. Virginia) Santhu (455384918158)

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Nminds-PublicSN-1
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az1 (us-east-1a)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.100.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional	
Name	Nminds-PublicSN-1	Remove

Add new tag

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSubnet:VpcId=vpc-0381f64ef8c59f6cc

Search [Alt+S]

United States (N. Virginia) Santhu (455384918158)

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Nminds-PrivateSN-1
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az2 (us-east-1b)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.200.0/24 256 IPs

▼ **Tags - optional**

Key	Value - optional	
Name	Nminds-PrivateSN-1	Remove

Add new tag

CloudShell Feedback Console Mobile App

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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#igws:

Search [Alt+S]

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VPC > Internet gateways

VPC dashboard < Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

- Network ACLs
- Security groups

PrivateLink and Lattice

Internet gateways (2) Info

Find internet gateways by attribute or tag

☐	Name	Internet gateway ID	State	VPC ID	Owner
☐	Default-IGW	igw-0e84d2986caff5c04	Attached	vpc-0490cf8ae8d826f24 Default-VPC	455384918158
☐	Nminds-NV-IGW	igw-0b56b658d5c447cd0	Attached	vpc-0381f64ef8c59f6cc Nminds-NV-VPC	455384918158

Select an internet gateway above

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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:

Search [Alt+S]

United States (N. Virginia) Santhu (455384918158)

VPC > Route tables

VPC dashboard < Filter by VPC

AWS Global View

Virtual private cloud

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Security

- Network ACLs
- Security groups

PrivateLink and Lattice

Route tables (1/3) Info

Find route tables by attribute or tag

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Public-RT	rtb-01a6db27665cb7c1e	-	-	Yes	vpc-0381f64ef8c59f6cc
☐	Default-RT	rtb-0480225f900d86226	-	-	Yes	vpc-0490cf8ae8d826f24
☐	Private-RT	rtb-014ba81d4b5123b82	-	-	No	vpc-0381f64ef8c59f6cc

rtb-01a6db27665cb7c1e / Public-RT

Details Routes Subnet associations Edge associations Route propagation Tags Related resources

Details

Route table ID
rtb-01a6db27665cb7c1e

VPC
vpc-0381f64ef8c59f6cc | Nminds-NV-VPC

Main
Yes

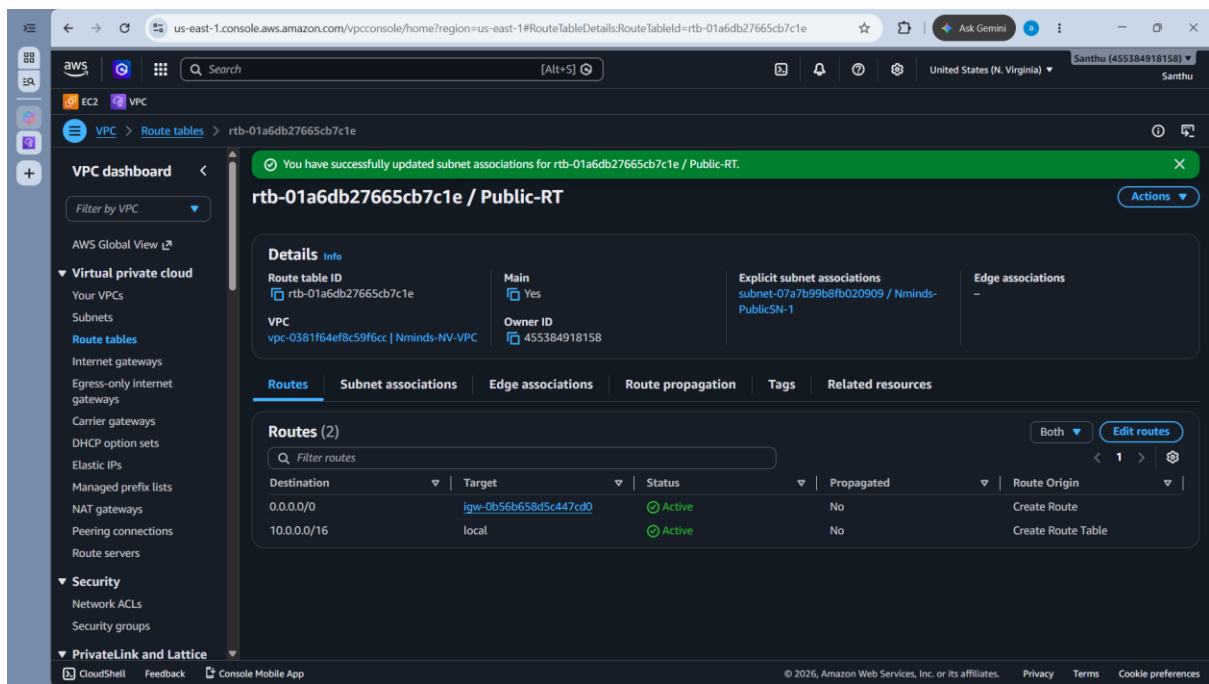
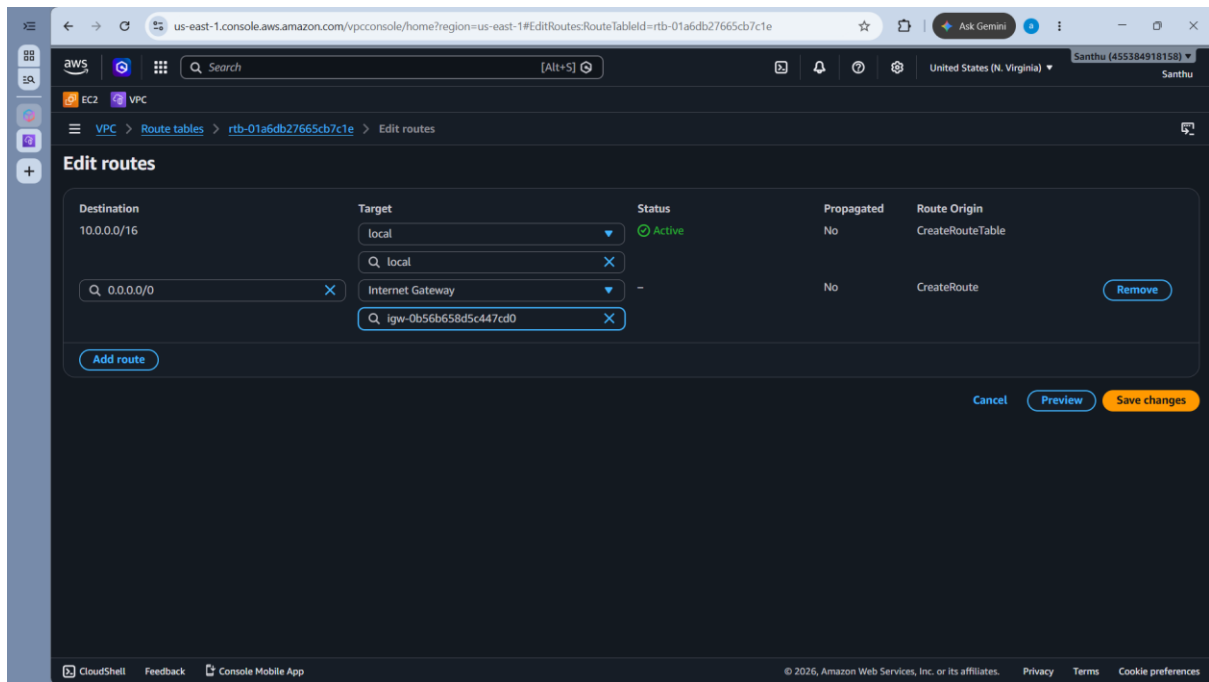
Owner ID
455384918158

Explicit subnet associations
-

Edge associations
-

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us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#EditRouteTableSubnetAssociations.RouteTableId=rtb-014ba81d...

EC2 VPC

VPC > Route tables > rtb-014ba81d4b5123b82 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Nminds-PrivateSN-1	subnet-Oa3b243463227f782	10.0.200.0/24	-	Main (rtb-01a6db27665cb7c1e / Publi...
☐	Nminds-PublicSN-1	subnet-07a7b99b8fb020909	10.0.100.0/24	-	rtb-01a6db27665cb7c1e / Public-RT

Selected subnets

subnet-Oa3b243463227f782 / Nminds-PrivateSN-1

Cancel Save associations

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us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSecurityGroup.VpcId=vpc-0381f64ef8c59f6cc

EC2 VPC

VPC > Security Groups > Create security group

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Nminds-PublicSN-SG

Name cannot be edited after creation.

Description [Info](#)

allow

VPC [Info](#)

vpc-0381f64ef8c59f6cc (Nminds-NV-VPC)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
SSH	TCP	22	Anyw...	0.0.0.0/0	Delete
				0.0.0.0/0	X
HTTP	TCP	80	Anyw...	0.0.0.0/0	Delete
				0.0.0.0/0	X

Add rule

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us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#SecurityGroupgroupId=sg-02fef8277fc27a92e

Security group (sg-02fef8277fc27a92e | Nminds-PublicSN-SG) was created successfully

sg-02fef8277fc27a92e - Nminds-PublicSN-SG

Details

Security group name Nminds-PublicSN-SG	Security group ID sg-02fef8277fc27a92e	Description allow	VPC ID vpc-0381f64ef8c59f6cc
Owner 455384918158	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0537bd5cb90daa6c6	IPv4	SSH	TCP	22
-	sgr-0965838b418ba76ee	IPv4	HTTP	TCP	80

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSecurityGroup:VpcId=vpc-0381f64ef8c59f6cc

Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name: Nminds-PrivateSN-SG

Description: allow

VPC: vpc-0381f64ef8c59f6cc (Nminds-NV-VPC)

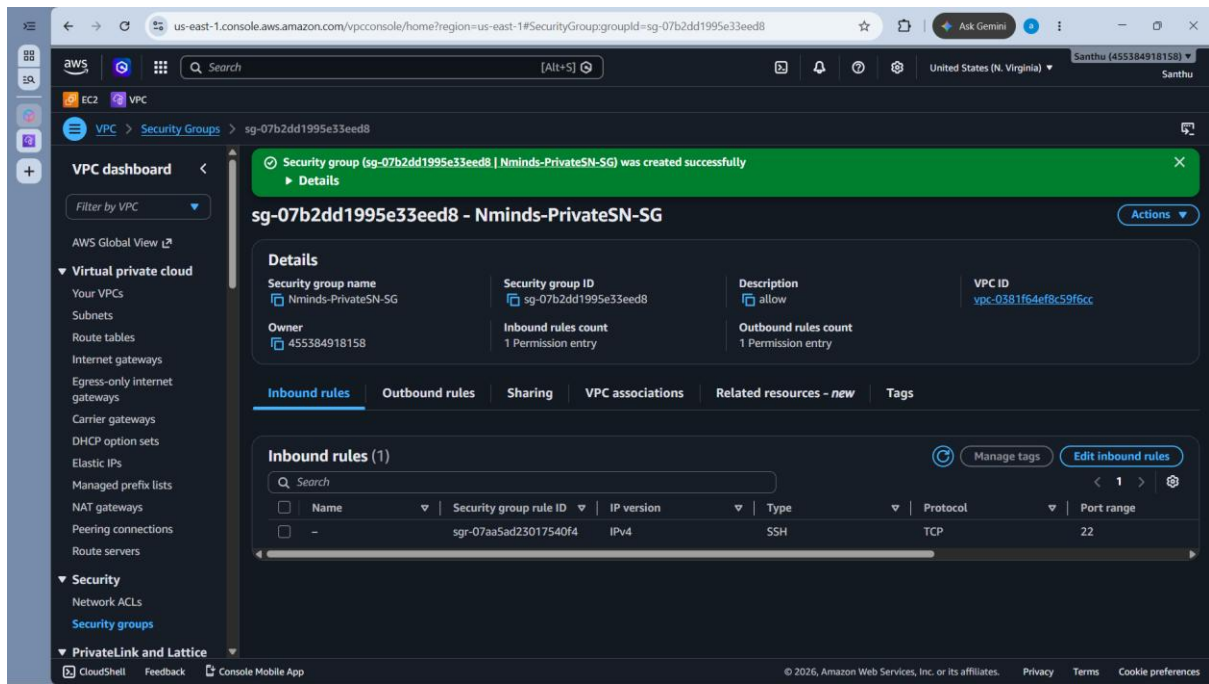
Inbound rules

Type	Protocol	Port range	Source	Description - optional
SSH	TCP	22	Anyw... 0.0.0.0/0	

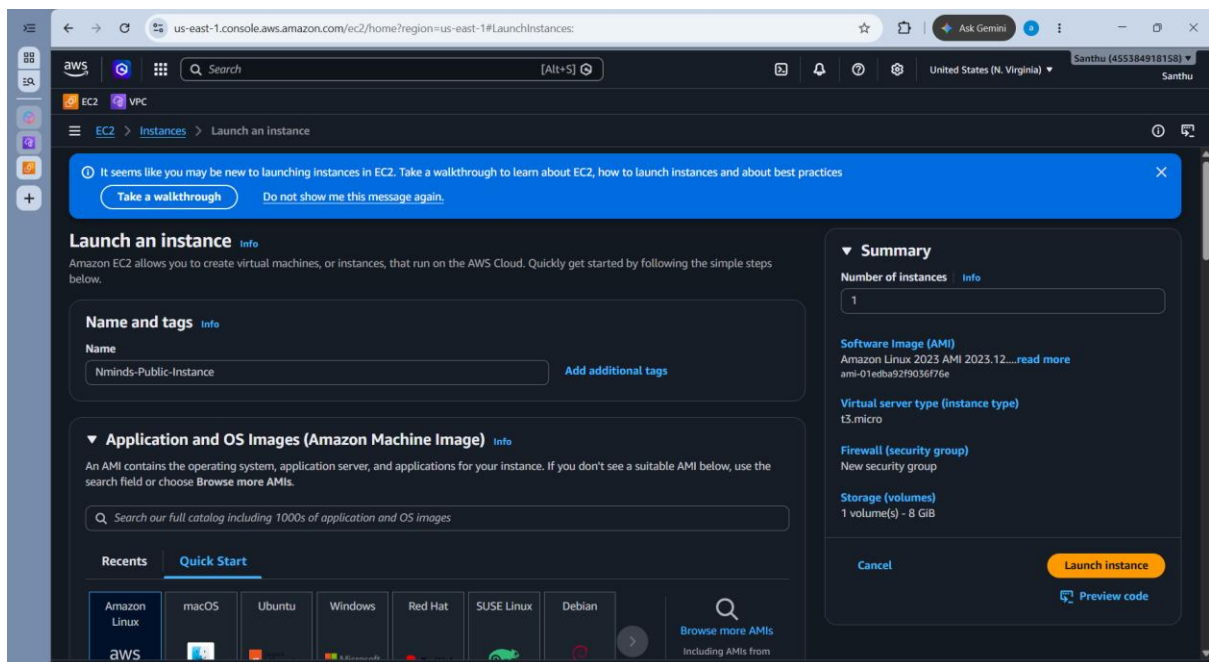
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules





Task2: Create Linux Instance & Install Apache Webserver (EC2) on newly created public subnet:



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:

EC2 > Instances > Launch an instance

Network settings

VPC - required

vpc-0381f64ef8c59f6cc (Nminds-NV-VPC)

Subnet

subnet-07a7b99b8fb020909 Nminds-PublicSN-1

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Common security groups

Select security groups

Nminds-PublicSN-SG sg-02fef8277fc27a92e

Advanced network configuration

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...read more

ami-01edba92f9036f76e

Virtual server type (instance type)

t3.micro

Firewall (security group)

Nminds-PublicSN-SG

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch Instance

Preview code

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:

EC2 > Instances

Instances (1/1)

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Nminds-Public...	i-059f615fd3c480abb	Running	t3.micro	Initializing	us-east-1a	-

i-059f615fd3c480abb (Nminds-Public-Instance)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-059f615fd3c480abb

Public IPv4 address

100.56.237.151 | open address

Private IPv4 addresses

10.0.100.157

IPv6 address

-

Instance state

Running

Public DNS

-



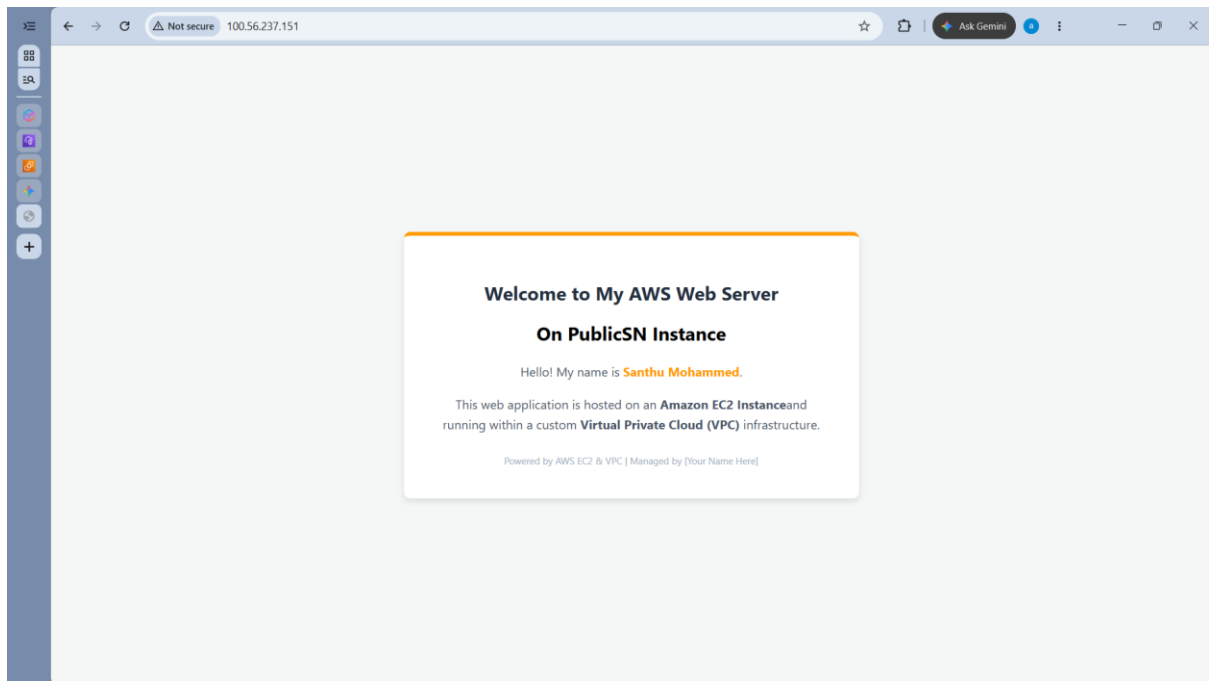

```

root@ip-10-0-100-157:/var/www
    font-size: 16px;
    line-height: 1.6;
  }
  .highlight {
    font-weight: bold;
    color: #ff9900;
  }
  .footer {
    margin-top: 25px;
    font-size: 12px;
    color: #a0aec0;
  }
}
</style>
</head>
<body>

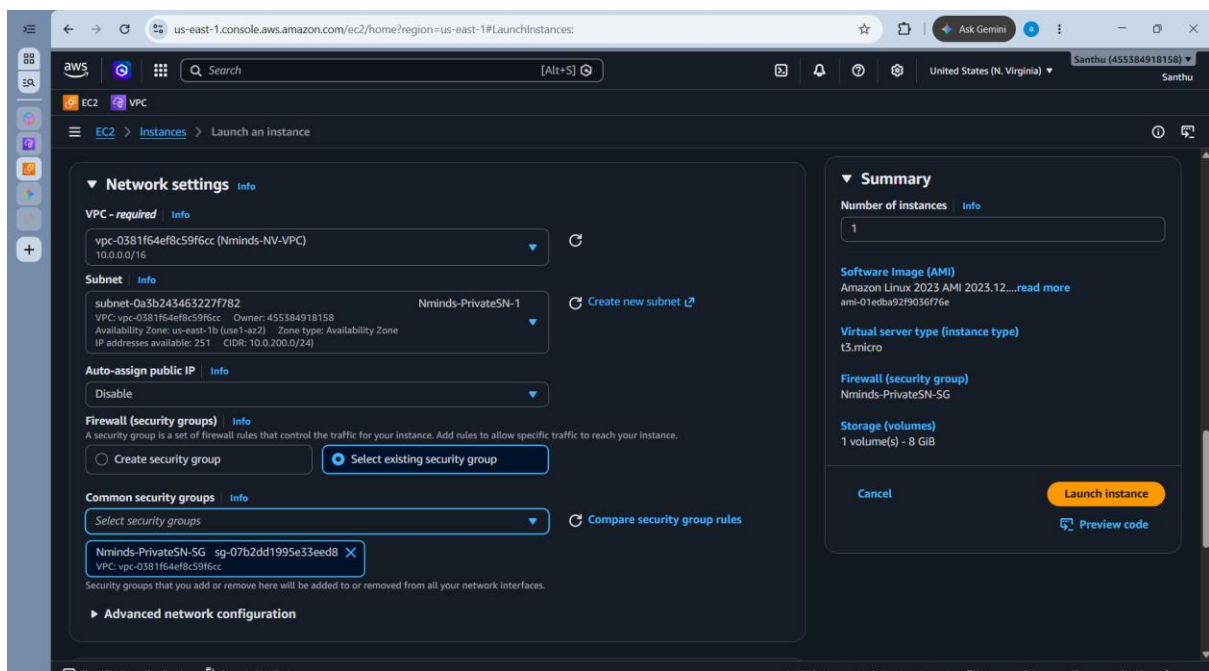
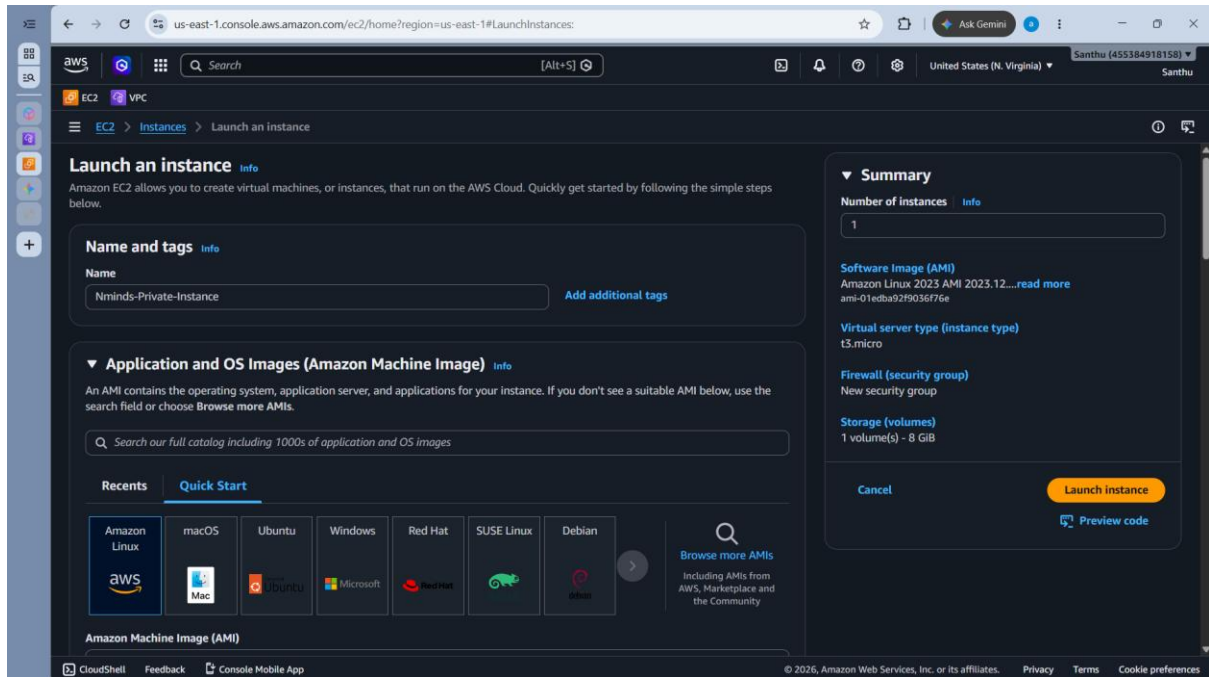
  <div class="card">
    <h1>Welcome to My AWS Web Server</h1>
    <h2>On PublicSN Instance</h2>
    <p>Hello! My name is <span class="highlight">Santhu Mohammed</span>.</p>
    <p>This web application is hosted on an <strong>Amazon EC2 Instance</strong>and running within a custom <strong>
Virtual Private Cloud (VPC)</strong> infrastructure.</p>
    <div class="footer">
      Powered by AWS EC2 & VPC | Managed by [Your Name Here]
    </div>
  </div>

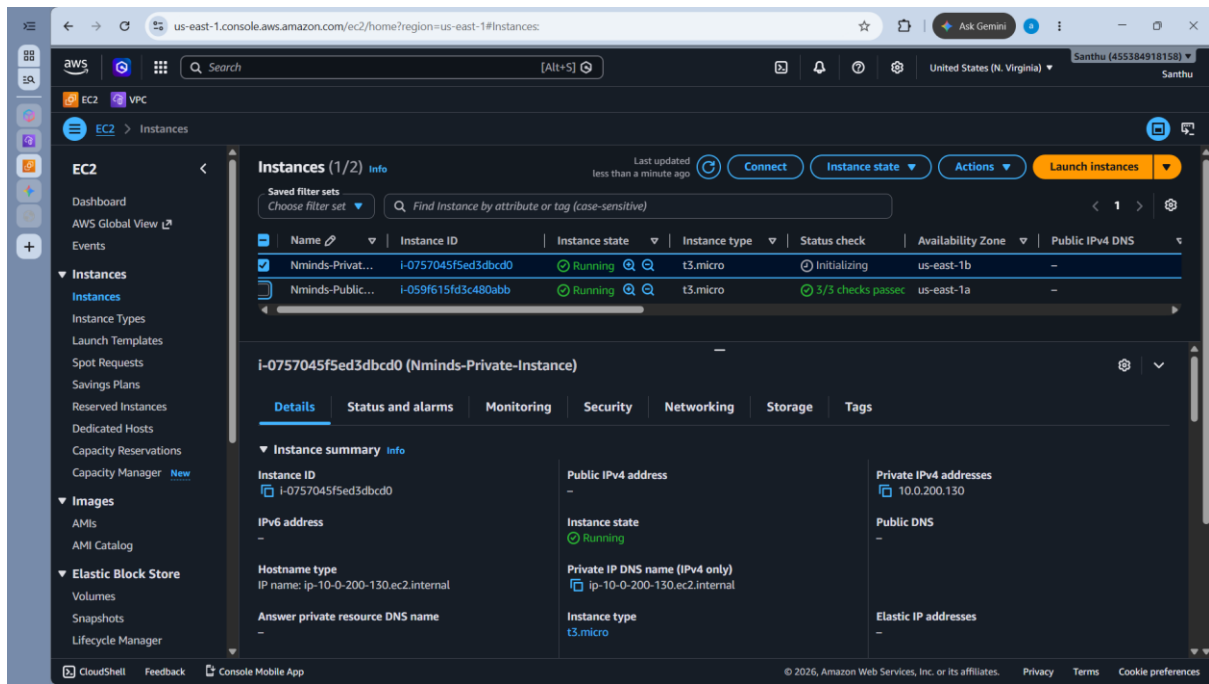
</body>
</html>

```

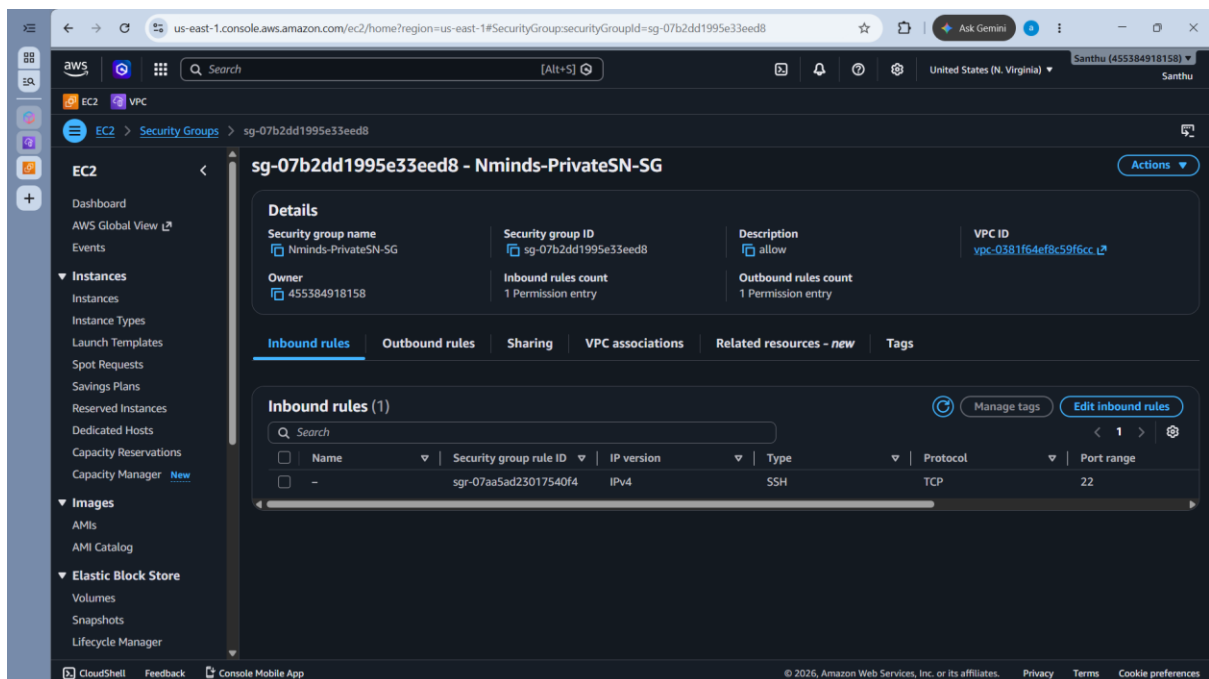


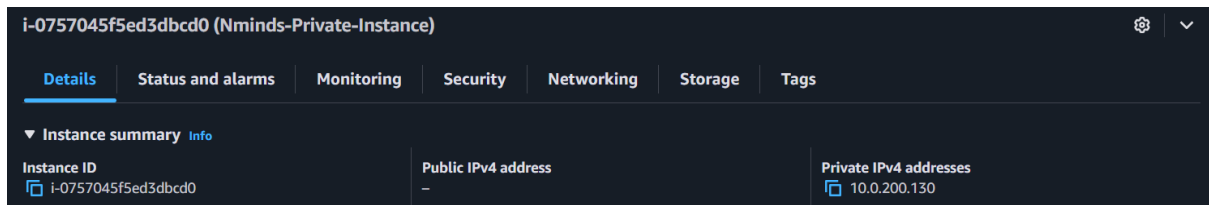
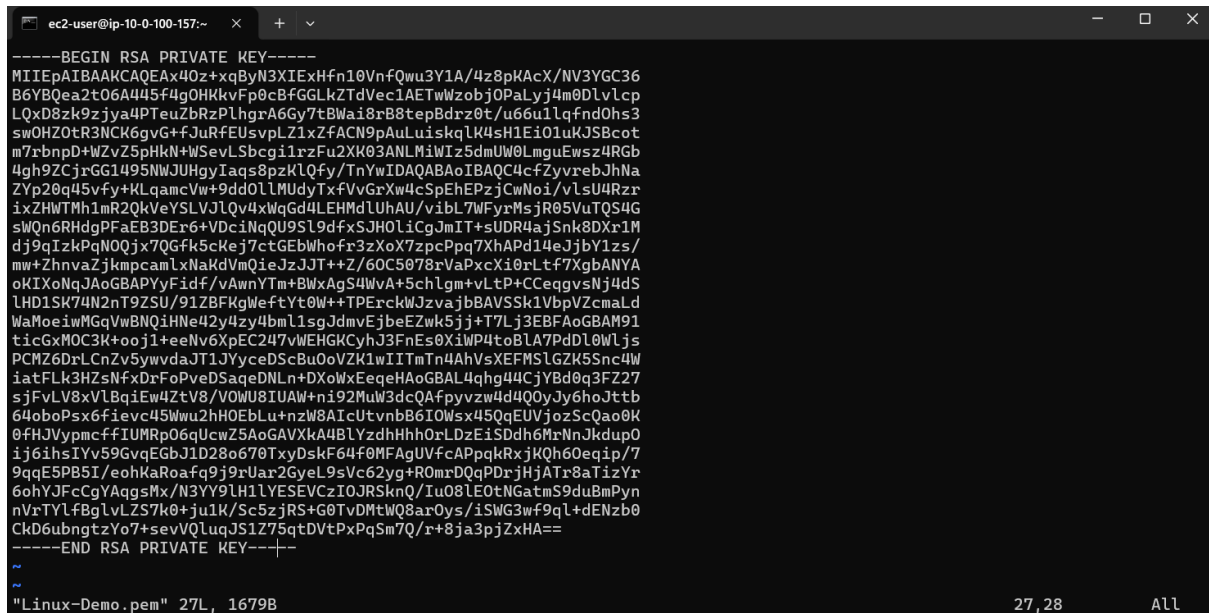
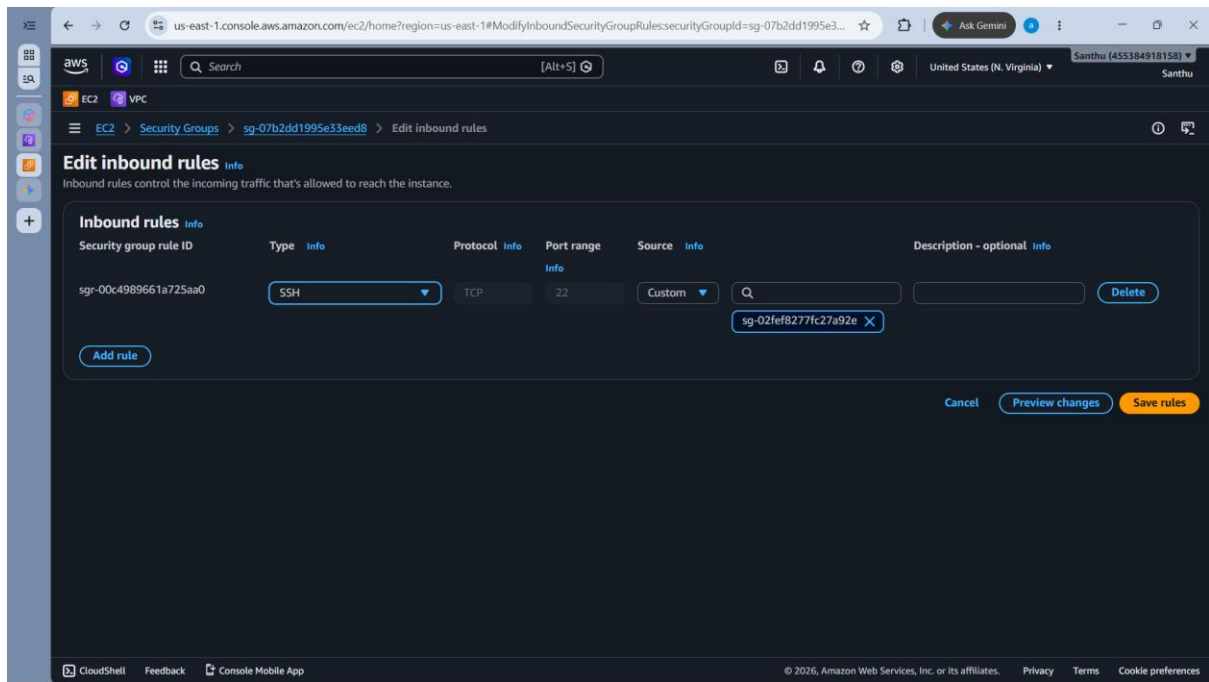
Task3: Create Linux Instance (EC2) on newly created private subnet:





Task4: The Private subnet Linux machine only able to access by public subnet instance only, how to configure for this scenario with execution steps





```
root@ip-10-0-100-157:~  
[ec2-user@ip-10-0-100-157 ~]$ sudo -i  
[root@ip-10-0-100-157 ~]# vi Linux-Demo.pem  
[root@ip-10-0-100-157 ~]# , 1679B written  
[root@ip-10-0-100-157 ~]# ssh -i Linux-Demo.pem ec2-user@10.0.200.130  
The authenticity of host '10.0.200.130 (10.0.200.130)' can't be established.  
ED25519 key fingerprint is SHA256:arV0vLZNDm+zcAWsx4n7NsG50pwBjM5WEEok+bj3wuc.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.200.130' (ED25519) to the list of known hosts.  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
@ WARNING: UNPROTECTED PRIVATE KEY FILE! @  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
Permissions 0644 for 'Linux-Demo.pem' are too open.  
It is required that your private key files are NOT accessible by others.  
This private key will be ignored.  
Load key "Linux-Demo.pem": bad permissions  
ec2-user@10.0.200.130: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
[root@ip-10-0-100-157 ~]# |
```

```
root@ip-10-0-100-157:~  
[ec2-user@ip-10-0-100-157 ~]$ sudo -i  
[root@ip-10-0-100-157 ~]# vi Linux-Demo.pem  
[root@ip-10-0-100-157 ~]# , 1679B written  
[root@ip-10-0-100-157 ~]# ssh -i Linux-Demo.pem ec2-user@10.0.200.130  
The authenticity of host '10.0.200.130 (10.0.200.130)' can't be established.  
ED25519 key fingerprint is SHA256:arV0vLZNDm+zcAWsx4n7NsG50pwBjM5WEEok+bj3wuc.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '10.0.200.130' (ED25519) to the list of known hosts.  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
@ WARNING: UNPROTECTED PRIVATE KEY FILE! @  
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@  
Permissions 0644 for 'Linux-Demo.pem' are too open.  
It is required that your private key files are NOT accessible by others.  
This private key will be ignored.  
Load key "Linux-Demo.pem": bad permissions  
ec2-user@10.0.200.130: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
[root@ip-10-0-100-157 ~]# ls -ld Linux-Demo.pem  
-rw-r--r--. 1 root root 1679 Jul 14 15:53 Linux-Demo.pem  
[root@ip-10-0-100-157 ~]# chmod 400 Linux-Demo.pem  
[root@ip-10-0-100-157 ~]# ls -ld Linux-Demo.pem  
-r-----. 1 root root 1679 Jul 14 15:53 Linux-Demo.pem  
[root@ip-10-0-100-157 ~]# |
```



